Geophagic Use of Calabash Chalk on Sperm Characteristics and Testicular Histoarchitecture of Adult Wistar Rats

[*1]Odiase D.E., [1]Maduakolam J.D., [2]Zekere C.S.

[1]Department of Anatomy, School of Basic Medical Sciences, College of Medical Sciences, University of Benin, Edo State, Nigeria.
[2]Department of Obstetrics and gynaecology, Igbinedion University, Okada, Edo State, Nigeria

ABSTRACT

Calabash chalk is composed of aluminium silicate hydroxide and belongs to the Kaolin clay family, naturally occurring mineral and it is derived from fossilized seashells. It is composed of several elements and also constitutes major remedy for morning sickness. The aim of this study is to evaluate the effects of calabash chalk on sperm characteristics and the histology of the testis of adult Wistar rats. Twenty (20) matured Wistar rats were randomly assigned into four groups of five rats each (A-D). Group A served as the control while group B, C and D served as the test groups and were treated with 100mg/kg, 200mg/kg and 400mg/kg of calabash chalk for 8 weeks. Semen were collected for sperm parameters analysis and the testes harvested for histological processing. There was no significant change ($P>0.05$) in the sperm parameters, luminal contents and histo-architecture of the testes of the treatment groups compare with the control group. These findings concluded that calabash chalk has no significant deleterious effects on the sperm content and testicular histoarchitecture of adult Wistar rats.

Keywords: Calabash chalk, Sperm, Semen analysis, Testes, Histology.

***Correspondence: Tel.:** 08067106187.**; E-mail:** daniel.odiase@uniben.edu

1. INTRODUCTION

Calabash chalk is mainly composed of aluminium silicate hydroxide- $Al_2Si_2O_5(OH)_4$ and belongs to the Kaolin clay family [1]. Calabash chalk is a geophagic element which are traditionally been consumed by humans (particularly pregnant women) as a remedy for nausea and vomiting during pregnancy. The powdered form may also be used as face powder and as antiperspirant [2]. It is available in different forms such as powders, molded shapes, and blocks and it comes in two varieties: salted and unsalted. Calabash chalk is also known as Nzu by the Ibos, Ndom by the Efik/Ibibios, and Eko in some sections of Benin in Nigeria [3].

It is widely distributed in Nigeria and other Sub-Saharan African regions, when in contact with digestive fluids, it may cause clinical or sub-clinical harmful effects in humans [4]. Research has demonstrated that it causes hepatic sinusoidal enlargements as well as gastrointestinal damage and anaemia in vulnerable population. The presence of elements such as lead and arsenic acid in Calabash chalk have been linked to nerve and brain damage resulting in learning and behavioral dysfunctions in both animals and humans. Also it has been shown that prolonged intake of Calabash chalk increases anxiety and pain perception in mice [1, 5].

Owhorji posited that there are safety issues regarding the consumption of Calabash chalk, which is prevalent in many African and Asian communities [6]. Calabash chalk is a geophagic substance, commonly consumed for pleasure in some West African countries, and most pregnant women seek it for sickness remedy [1, 4, 7, 8]. However, the practice of geophagia may expose the consumer to

harmful chemicals and parasites contained in the chalk consumed [5].

Obidike enumerated some reasons to avoid consuming calabash chalk to include; having no nutritional value as it lacks carbohydrate, protein, dietary fiber, fats, and vitamins, being addictive, it may contain organic pollutants that may be harmful to the body [9]. Also, consuming calabash chalk during pregnancy was discovered to have a harmful effect because of its lead and arsenic concentration resulting in pregnancy losses, stillbirth and foetal growth retardation [10]. Despite this information on it non-nutritive value, and effect on pregnancy and foetal health, there is paucity of literature on male reproductive function. It is therefore the aim of this study to evaluate the effects of calabash chalk on sperm characteristics and the histology of the testis of adult Wistar rats.

2. MATERIALS AND METHODS

2.1. Collection and identification of materials: Calabash chalk was obtained at Utezi community in Egor Local Government area of Edo state, Nigeria. The substance was identified at the Department of Chemistry, University of Benin, Edo State, Nigeria.

2.2. Experimental animals: Twenty Wistar rats weighing between 180g and 200g were procured and housed in the Department of Anatomy Animal House unit for a period of two weeks for acclimatization. They were kept in clean, well-ventilated aluminium cage at room temperature.

2.3. Preparation of test material: Calabash chalk, which is a solid mast, was hand-ground to powder and then placed in a basin of water until entirely dissolved. It was then carefully poured into a storage container before being administered to the animals.

2.4. Determination of dose: LD50 of calabash chalk was determined to be greater than 5000mg/kg and this was used to set the dose for administration [11].

2.5. Experimental design: Twenty (20) rats were divided into four groups (n = 5). Group A served as the control group while groups B, C and D were the test groups and were administered the dissolved calabash chalk at doses of 100mg/kg, 200mg/kg and 400mg/kg respectively for 8 weeks. All animals received their regular feed and water *ab libitum* during the experiments.

2.6. Sacrifice of animals: At the end of the treatments, rats were sacrificed under moderate anaesthesia with chloroform. The testes and epididymis were harvested and weighed. Semen were extracted from the vas deferens and analysed for viability, morphology, and count using a counting chamber and microscope. The testes were preserved in Bouin's solution and prepared for histology processing.

2.7. Data analysis: The data obtained were expressed as mean ± SEM and analysed for statistical significance at $p<0.05$ using the using one-way ANOVA of Statistical Package for Social Sciences and presented using suitable tables and figures.

3. RESULTS

The results showed no-significant different in the initial and final body weight of rats (Table 1). Similarly, there was no significant different in the testicular weight and testiculosomatic index between groups, however, they were relatively higher in the calabash chalk treated groups with greatest effect in the 100mg/kg treated group (table 2). Figure 1 compares the semen charateristics following calabash chalk ingestion and control. Pecentage progressive motility, non-progressive motility, and immotile sperm cells were not significantly difference compared with control. Also, there were normal testicular architecture with normal seminiferous tubules (A) lined by normal spermatogenic series (B).

Table1. Initial and final body weight of rats in treatment groups compared with control

GROUPS	Initial body weight	Final body weight	P-VALUE
Control	189.00±2.00	175.33±3.53	0.120
100mg/kg Calabash chalk extract	175.60±7.01	182.80±3.77	0.513
200mg/kg Calabash chalk extract	170.00±12.50	175.00±18.45	0.863
400mg/kg Calabash chalk extract	173.00±8.44	182.75±17.20	0.432

Table 2: Testicular weight and testiculosomatic indices of rats in treatment groups compared with the control.

	Control	Calabash chalk treated			P-value
		100mg/kg	200mg/kg	400mg/kg	
Testicular weight (g)	1.02±0.04	1.26±0.09	1.08±0.04	1.08±0.13	0.287
Testiculosomatic index (%)	0.58±0.01	0.69±0.04	0.63±0.04	0.59±0.06	0.327

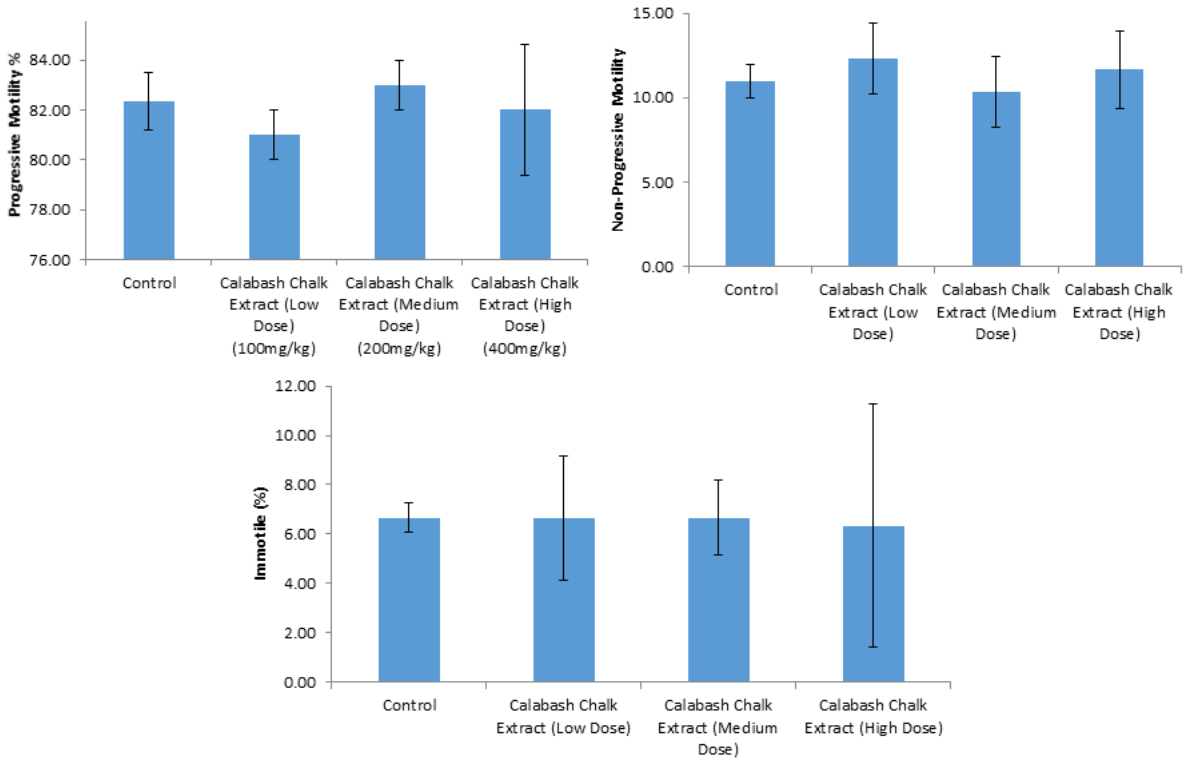


Figure 1. Percentage progressive and non-progressive motility and immotile sperm after the 8 weeks of calabash chalk ingestion.

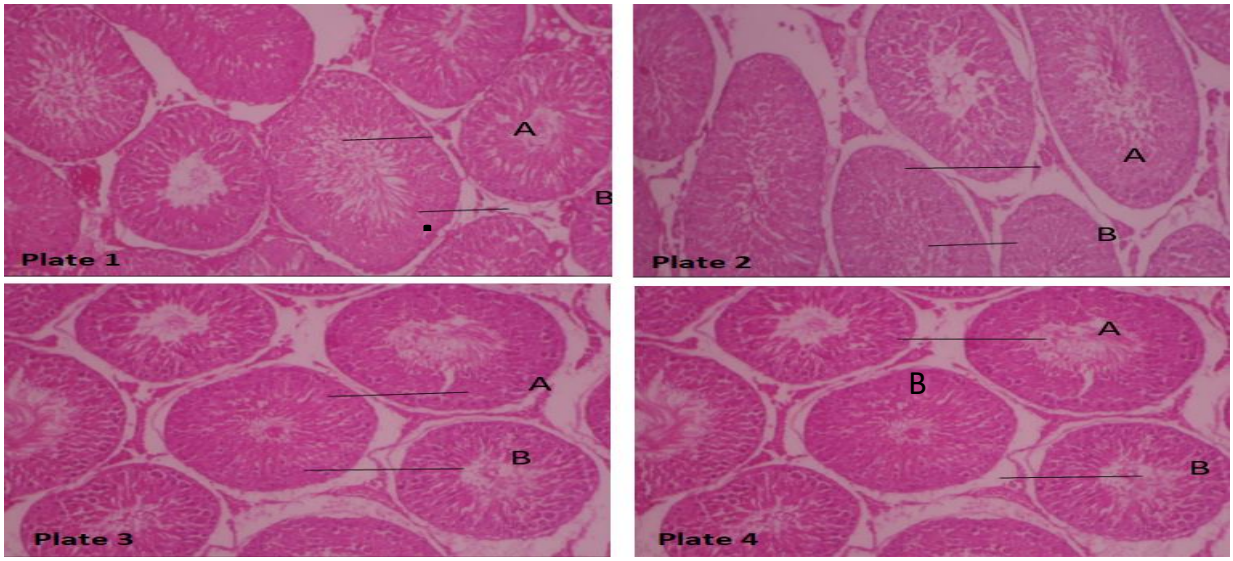


Figure 2. Plates 1, 2, 3 and 4 are photomicrographs of rat testes of control, low, medium and high doses treated calabash chalk groups respectively (H&E x 100).

4. DISCUSSION

Increase in the final body weight was observed in all the groups when compared to the initial body weight. Although most of these increases were not statistically significant, however, it has been argued that increase or decrease in either absolute or relative weight of an animal after administering a chemical or drug is an indication of the toxic effect of that chemical.

In other parameters that were assessed including testicular weight, progressive motility, non-progressive motility, and immotile sperm cells, no statistically significant difference s($P<0.05$) was observed across the groups when compared with the control group.

In the histological slide of this study, the testes of rats that were administered low, medium and high doses of calabash chalk showed normal testicular architecture, with seminiferous tubules lined by spermatogenic cells at different levels of maturation, mild Leydig cell hyperplasia were observed in the testes across all groups of animals administered with calabash chalk when compared with control. These cells are responsible for the production of testosterone in human males [12]. The histology also shows increased luminal diameter of the seminiferous tubule in groups of animals that were treated with Calabash Chalk when compared with the control group. Although, calabash chalk is believed to contain different element such as lead which might be harmful to humans. However, the result of this study did not reflect any significant alteration in the histology of the testes across all the treated groups. We could deduced that the concentration of such elements in the chalk may not be significant to the extent of been harmful to the rapidly proliferating cells of the testes. This contradicts the earlier result by Ekong et al which claimed that calabash chalk could be a source of pollutants that could be hazardous to the human body [5].

5. CONCLUSION

Findings from this study indicate that oral consumption of Calabash chalk has no

deleterious effect on the testes of an adult male Wistar rat. This is contrary to earlier assertion by some researchers that it a source of pollutants that may be harmful to humans.

REFERENCES

1. Dean, J.R., Deary, M.E., Gbefa, B.K., Scott, W.C., (2004). Characterization and analysis of persistent organic pollutants and major, minor and trace elements in calabash stone. *Chemosphere* 57 (1), 21–25.
2. Popoola OE, Bisi-Johnson MA, Abiodun A and Ibeh OS. (2013). Heavy metal content and antimicrobial activities of some naturally occurring facial cosmetics in Nigeria. *Ife Journal of Science*, 15(3), 637-645.
3. Ekong MB, Akpantah AO, Ibok OS, Eluwa MA and Ekanem TB. (2008). Differentia effect of calabash chalk on the histology of the liver of adult Wistar rats. *The Internet Journal of Health,* 8(2), 1-4.
4. Abrahams PW, Follansbee MH, Hunt A, Smith B, Wragg J (2006). Iron nutrition and possible lead toxicity: An appraisal of geophagy undertaken by pregnant women of UK Asian communities. *Applied Geochemistry*. 21, 98-108.
5. Ekong MB, John EE, Mbadugha CC, Bassey EI and Ekanem TB. (2012). Effects of calabash chalk on the histomorphology of the gastro-oesophageal tract of growing Wistar rats. *The Malaysian Journal of Medical Sciences*, 19(1), 30-35.
6. Owhorji B, Okon U, Nwankwo A, Osim E. (2019). Chronic consumption of calabash chalk diet impairs locomotor activities and social behaviour in Swiss white Cd-1 mice. Heliyon 5 (2019) e01848
7. Akpantah, A.O., Ibok, O.S., Ekong, M.B., Eluwa, M.A. and Ekanem, T.B., 2010. The effect of calabash chalk on some hematological parameters in female adult Wistar rats. *Turkish Journal of Hematology*, *27*(3), pp.177-181.
8. Moses, B.E., Emma, E.J., Christopher, C.M., Enobong, B. and Theresa, B.E., (2012). Effect of calabash chalk on the histomorphology of the gastro-oesophageal tract of growing Wistar rats. *The Malaysian Journal of Medical Sciences: MJMS*, *19*(1): 30.
9. Obidike J. (2021): Top ten reasons why you should quit calabash chalk (Nzu). *Health A-Z, Health news* https://www.healthfulwonders.com/10-reasons-to-quit-eating-calabash-chalk-nzu/
10. Aprioku, J. S., & Ogwo-Ude, E. M. (2018). Gestational toxicity of Calabash chalk (Nzu) in Wistar rats. *International Journal of Applied and Basic Medical Research*, *8*(4), 249.
11. Ekanem, T. B., Ekong, M. B., Eluwa, M. A., Igiri, A. O., & Osim, E. E. (2015). Maternal geophagy of calabash chalk on foetal cerebral cortex histomorphology. Malaysian *Journal of Medical Sciences*, 22(4), 17–22.
12. Mohammed, A, Mohammed, A, Serah, A, Suhair, AQ, Seleh, A, and Tariq SQ 2022: Bilateral Leydig cell Hyperplasia Presented Incidentally: A Case Report. *International Journal of Surgery Case Report* 90:106-733.